

ORIGINAL ARTICLE

Ultrasonographic Percutaneous Anatomy of the Atlanto-Occipital Region and Indirect Ultrasound-Guided Cisternal Puncture in the Dog and the Cat

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With 5 figures

Received January 2013; accepted for publication February 2014

doi: 10.1111/ah.12113

Summary

Cisternal puncture in dogs and cats is commonly carried out. This article describes the percutaneous ultrasound anatomy of the cisternal region in the dog and the cat and an indirect technique for ultrasound-guided cisternal puncture. Ultrasound images obtained *ex vivo* and *in vivo* were compared with anatomic sections and used to identify the landmarks for ultrasound-guided cisternal puncture. The ultrasound-guided procedure was established in cadavers and then applied *in vivo* in seven dogs and two cats. The anatomic landmarks for the ultrasound-guided puncture are the cisterna magna, the spinal cord, the two occipital condyles on transverse images, the external occipital crest and the dorsal arch of the first cervical vertebra on longitudinal images. Using these ultrasound anatomic landmarks, an indirect ultrasound-guided technique for cisternal puncture is applicable in the dog and the cat.

Introduction

The introduction of a spinal needle in the cisterna magna (CM) is commonly carried out for cerebrospinal fluid (CSF) collection (Oliver et al., 1997; Ducoté and Dewey, 2003) or to perform myelography (Adams, 1982) or computed tomography (CT) myelography (Shimizu et al., 2009; Robertson and Thrall, 2011) in dogs. This procedure is considered technically easier than the lumbar puncture (Oliver et al., 1997) but can be responsible for iatrogenic injury (Lujan Feliu-Pascual et al., 2008). The anatomic landmarks for the cisternal tap are the external occipital protuberance and the spinous process of the axis (C2) for the midline and the cranial border of the transverse process (wings) of the atlas (C1) (Ducoté and Dewey, 2003; Oliver et al., 1997; Tipold, 2003; Elias and Brown, 2008). These landmarks are usually identified by palpation, and introduction of the spinal needle is

performed at the intersection of the midline and a line connecting the wings of the C1 (Elias and Brown, 2008). As in human medicine for cervical myelography (Orrison et al., 1983; Yousem and Gujar, 2009), fluoroscopic guidance can be used to puncture the CM. Ultrasound (US) guidance is used in human medicine to guide a needle for regional anaesthesia (Marhofer et al., 2005), to insert an epidural catheter in children (Rapp et al., 2005; Willschke et al., 2006) and to puncture the lumbar subarachnoid space (Peterson and Abele, 2005; Ferre and Sweeney, 2007). US guidance is considered to reduce the risks of the procedure and to increase operator's confidence.

US guidance has been described for CSF collection at the CM (Audigié et al., 2004), the lumbosacral space (Aleman et al., 2007) and more recently between C1 and C2 (Pease et al., 2012) in horses. An US-guided technique for lumbar subarachnoid space puncture has been described in dogs (Etienne et al., 2010). Sonographic

assessment of the craniocervical junction has been used to detect congenital malformations in humans (Braun et al., 1983; Cramer et al., 1986) without exposure to ionizing radiation. Recently, in veterinary medicine, the ultrasonographic appearance of the canine CM and the Chiari-like malformation has been described in Cavalier King Charles Spaniels dogs (Schmidt et al., 2008).

The aims of this study are to describe the US percutaneous anatomy of the atlanto-occipital region in dogs and cats and to illustrate the use of an indirect US-guided technique (Nyland et al., 2002) (transducer removed after determination of the angle and depth of needle insertion from the US image) for cisternal puncture.

Materials and Methods

The study was conducted in four stages. All procedures of this study were performed in accordance with the Ethical Committee of the Faculty of Veterinary Medicine, Liège University.

Stage 1: cadaver study to establish detailed sonographic anatomy in comparison with cross anatomical sections

Ex vivo US images were obtained by examining the atlanto-occipital region of cadavers of 5 dogs euthanized (medetomidine 15–30 µg/kg intravenously, 5 min before pentobarbital 20% (100–150 mg/kg) intravenously) for reasons unrelated to this study: a Bernese mountain dog, a Labrador Retriever, a Poodle, a Beagle and an American Staffordshire Terrier. The normal cross-sectional anatomy of the canine craniocervical region was determined using frozen anatomic sections obtained from two previous cadavers (a Beagle and an American Staffordshire Terrier). These sections were then compared with the US images to establish the US anatomy of the canine atlanto-occipital region.

Stage 2: in vivo study to examine the feasibility of atlanto-occipital region sonography in live dogs and cats

Five normal awake dogs of different morphology, weighing between 6 and 33 kg, were used. Two awake European cats, weighing 4 and 5 kg, were also used for this step.

Stage 3: cadaver study to test the indirect US-guided technique

Two dog cadavers euthanized for reasons unrelated to this study were used. The same protocol for euthanasia was used as in stage 1.

Stage 4: in vivo study to apply the US-guided technique on clinical patients

The US-guided technique for cisternal puncture was applied in clinical conditions in seven dogs and two cats undergoing CSF puncture or myelography. For this *in vivo* study, patients were under general anaesthesia: animals were pre-medicated with intravenous methadone (0.2 mg/kg) and, 10–20 min later, intravenous midazolam (0.3 mg/kg). For induction propofol (approximately 4 mg/kg) but dosed according to demand was administered intravenously. Anaesthesia was maintained using inhalational anaesthetics (isoflurane) with flow rate adapted appropriately by a dedicated anaesthesiologist. A rigid endotracheal tube was used to prevent airway collapse or obstruction. The animal's oxygen saturation was monitored by the anaesthetist throughout the procedure. The animals were fasted before anaesthesia. The skin was prepared aseptically after clipping. Alcohol was used instead of gel to improve skin coupling. The hands of the operator were covered with sterile gloves. The probe was disinfected with Tristel Duo® (Tristel Solutions Limited, Cambridgeshire, UK), a cleaning and sporicidal disinfectant for US probes.

Positioning and equipment

For all stages, US images were acquired with the dog in lateral recumbency with a maximal flexion of the neck and the atlanto-occipital joint (angle between neck and nose was approximately 50 degree on cadavers and anesthetized dogs). An additional precaution was taken to stages 3 and 4 where the nose of the dog was placed and held parallel to the table during the puncture. The skin was clipped, and US gel (stages 1, 2 and 3) or alcohol (stage 4) was used to improve skin/probe coupling. In each dog, *ex vivo* (stages 1 and 3) and *in vivo* (stage 2) US images were obtained independently by two operators. In stage 4, only one operator performed US images. For all studies, US images were obtained with an Esaote My Lab Class C machine (Esaote Group, Via A. Siffredi, 58, 16153 Genova Italy). For the anatomical studies (stages 1 and 2), it was equipped with either a linear 7.5 MHz and a microconvex 7.5 MHz transducer depending on size, amount of fat and conformation of the dog. The microconvex probe was more often used for the small dogs (e.g. Poodle), while the linear probe was more often used for the large dogs (e.g. Bernese mountain dog).

For indirect US guidance (stages 3 and 4), the linear 7.5 MHz transducer was used. Transverse and longitudinal (sagittal) images were obtained in all stages. Spinal needles with a bevel of Quincke (BD, Becton Dickinson S.A., S. Agustin del Guadalix, Madrid, Spain) were used.

Needle diameter was 22 gauge. The needle length required for the puncture is estimated by measuring the distance between the skin surface and the dorsal part of the CM on the US image. The length used was usually 38 mm, but a longer needle (90 mm) was sometimes necessary.

Technique for US guidance

The external occipital protuberance and the cranial border of the transverse processes (wings) of the C1 were palpated. The triangular space formed by these landmarks was visualized by US. The CM was located in transverse and longitudinal sonographic images. Transverse US images were obtained by orienting the probe perpendicular to an imaginary line linking the external occipital protuberance to the spinous process of C2 (the cervical vertebral column) and by positioning the space between occipital condyles in the centre of the image. To obtain longitudinal US images, the transducer was rotated 90° into a longitudinal plane and placed in the median plane of the cervical vertebral column. The US window visible between the occipital bone and C1 in the sagittal plane was placed in the centre of the image.

For indirect US guidance (stages 3 and 4), the linear probe was held by the operator who carried out the puncture (Fig. 1), while a second operator traced the lines on the skin with a marker. Transverse and longitudinal (sagittal) US images of the craniocervical junction were used to trace the landmarks for this procedure. The first step of indirect cisternal puncture consisted of the identification of the anatomic landmarks. Once occipital condyles with the CM in the centre had been identified on transverse images with the probe perpendicular to the skin, the CM was centred on the screen (Fig. 1a). Two marks with a felt-tip one on each side of the probe were drawn, on the clipped skin. Then, the probe kept perpendicular to the skin was rotated 90° to obtain a sagittal image (Fig. 1b). During this procedure, the skin was not moved to keep the correct first marks. The acoustic window between occiput and C1 containing the CM was centred on the screen. The distance between CM and the

skin surface was measured to estimate the depth at which the needle should be inserted to obtain CSF. Two new landmarks on each side of the probe were drawn with a felt-tip (Fig. 1b). The probe was taken off. Two lines linking the opposite marks were drawn. The intersection point of these two lines was the site of introduction of the spinal needle (Fig. 1c). This was placed perpendicular to the skin (the same angle probe/skin). The operator who carried out the puncture with one hand held the skin with the other hand to avoid moving the skin. The needle was advanced very slowly until CSF was reached and become visible in the hub. Once CSF was obtained, a sample collection tube was placed under the hub of the spinal needle to collect the drops of CSF and/or the syringe with the non-ionic contrast was connected if a myelogram was performed.

Results

Normal ultrasonographic images (Stages 1 and 2)

Below the skin, the echoic cervical fascia covering the hypo-echoic muscles located on each side of the midline was visualized. Transverse and longitudinal images of the atlanto-occipital region were obtained between the occipital condyles and the dorsal arch of C1.

The dorsal atlanto-occipital membrane (covered at its deep aspect by the dura mater) was easy to identify in all animals. It appeared as a thin (1–2 mm) echoic, slightly curved line: slightly convex on transverse images (Figs 2 and 3) and slightly concave on longitudinal images (Figs 4 and 5) (depending on the degree of flexion of the neck and the atlanto-occipital joint). It continued the hyperechoic, dorsal edge of the foramen magnum to the hyperechoic cranial border of the dorsal arch of the C1 (Evans, 1993) and marked the dorsal boundary of the CM. It was covered dorsally by the *rectus capitis dorsalis muscle*.

On transverse images (Figs 2 and 3), the two occipital condyles were seen as symmetrical, parasagittal bony structures giving an acoustic shadowing. The dorsal aspect

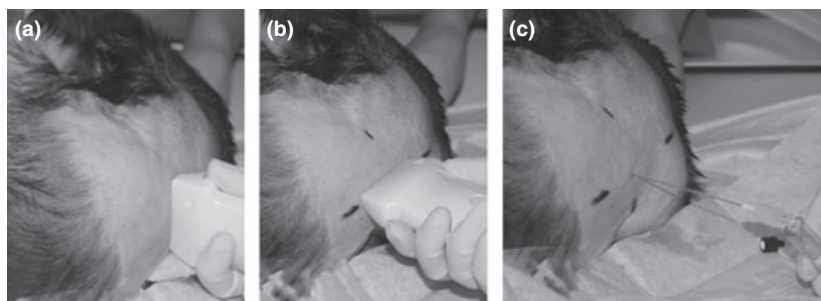


Fig. 1. Successive steps for indirect ultrasound-guided cisternal puncture. Identification of the ultrasonographic anatomic landmarks on transverse images (a) and longitudinal images with drawing marks on the skin (b) and introduction of the spinal needle (c).

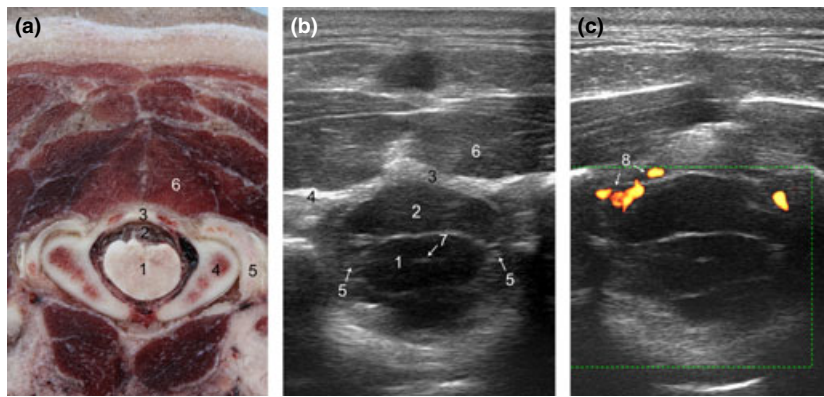


Fig. 2. Transverse anatomical section (a) and ultrasonographic images in B-mode (b) and with power Doppler (c) of the atlanto-occipital region of an American Staffordshire Terrier (a) and a crossed Shepherd dog (b&c), respectively. (a) 1, spinal cord; 2, *cisterna magna*; 3, dorsal atlanto-occipital membrane (with the dura mater); 4, occipital condyles; 5, cranial articular surfaces of the atlas (*fovea articularis cranialis*); 6, *rectus capitis dorsalis* muscle; (b) 1, spinal cord; 2, *cisterna magna*; 3, dorsal atlanto-occipital membrane (with the dura mater); 4, occipital condyles; 5, denticulate ligament; 6, *rectus capitis dorsalis* muscle; 7, central canal; (c) 8, dorsal branches of the spinal artery.

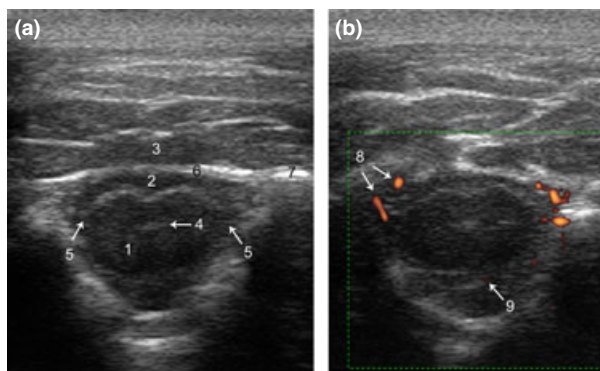


Fig. 3. Transverse ultrasonographic images in B-mode (a) and with power Doppler (b) of the atlanto-occipital region of an European cat. 1, spinal cord; 2, *cisterna magna*; 3, *rectus capitis dorsalis* muscle; 4, central canal; 5, denticulate ligament; 6, dorsal atlanto-occipital membrane (with the dura mater); 7, occipital condyles; 8, dorsal branches of the spinal artery; 9, basilar artery.

of the condyles appeared as a curved hyperechoic line. The occipital condyles were located close to the transducer surface in cats and small thin dogs (1–1.2 cm), while they were deeper in larger and fatter dogs. The basilar part of the occipital bone defined the ventral bony border of the CM. On longitudinal images in the median plane, the occipital condyles were obviously not seen.

Longitudinal images (Figs 4 and 5) were obtained between the occiput and C1. The external occipital crest was the cranial bony structure associated with an anechoic band due to acoustic shadowing on the median plane. It was visualized as a hyperechoic more or less slanting line depending on the animal species and flexion of the neck. In cats, this bony surface is straighter and

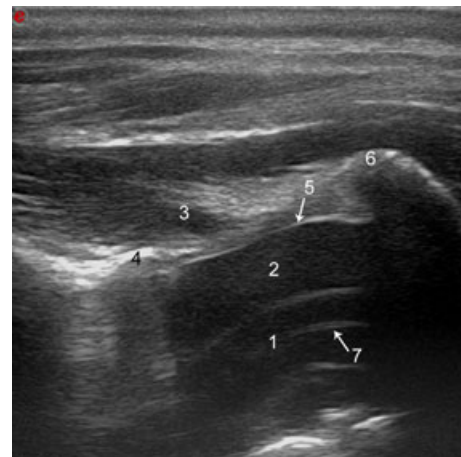


Fig. 4. Longitudinal ultrasonographic image of the atlanto-occipital region of a crossed Shepherd dog. 1, spinal cord; 2, *cisterna magna*; 3, *rectus capitis dorsalis* muscle; 4, external occipital crest; 5, dorsal atlanto-occipital membrane (with the dura mater); 6, dorsal arch of atlas; 7, central canal.

more oblique than in dogs (Fig. 5). The caudal bony structure with an acoustic shadow on the sagittal images was the dorsal arch of C1. It appeared as a hyperechoic slanting line.

The CM was visualized on sagittal images (Figs 4 and 5) between the external occipital crest (dorsal rim of the foramen magnum) and the dorsal arch of C1 and on transverse views between the occipital condyles. It appeared as an anechoic band dorsal to the spinal cord. On sagittal images, the spinal cord was visualized as a hypo-echoic structure outlined by two thin echoic lines representing the pia mater and with a visible central canal for US median slices. The central canal appeared as a very

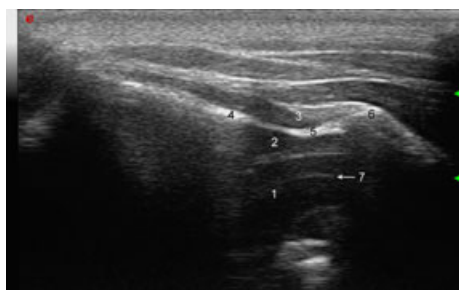


Fig. 5. Longitudinal ultrasonographic image of the atlanto-occipital region of an European cat. 1, spinal cord; 2, *cisterna magna*; 3, *rectus capitis dorsalis* muscle; 4, external occipital crest; 5, dorsal atlanto-occipital membrane (with the dura mater); 6, dorsal arch of atlas; 7, central canal.

thin echoic line (Fig. 5) or 2 very thin echoic and parallel lines (Fig. 4). The ventral portion of the CM was seen as an anechoic band thinner than the dorsal portion. On transverse views (Figs 2 and 3), the spinal cord appeared anechoic, ovoid (larger than thicker) and delineated by a hyperechoic line, representing the pia mater. The spinal cord was suspended in CSF within the subarachnoid space bilaterally along the lateral surface by the denticulate ligaments visualized as several hyperechoic, parallel and short structures. The central canal was visualized as a hyperechoic central spot.

The basilar artery was seen as a thin, pulsing vessel visualized ventral to the spinal cord from C1 cranially in dogs and cats (Fig. 4). The dorsal visible vessels, lying in the pia, were the dorsal branches of the spinal artery (Figs 2 and 3).

Indirect US-guided cisternal puncture (Stages 3 and 4)

On the 2 dog cadavers (stage 3), transverse and longitudinal images of the CM (Figs 2 and 4) were obtained with the probe placed in the two perpendicular positions illustrated in Fig. 1a,b. While inserting the needle at the intersection of the two lines traced based on these images (Fig. 1c), the CM was successfully reached.

Seven dogs and two cats underwent US-guided puncture (stage 4). In all animals, the first attempt to puncture the atlanto-occipital subarachnoid space was successful. Based on the macroscopic appearance, CSF was obtained without blood contamination in all patients. The procedure was considered easy and the estimation of depth of the CM increased the confidence of the operator during the puncture.

Discussion

The *ex vivo* part of this study was useful to compare US images with anatomic sections, to fully understand the

US anatomy of the region and to test the US-guided technique before its application *in vivo*. As reported in human patients (Peterson and Abele, 2005; Ferre and Sweeney, 2007; Stiffler et al., 2007) and in dogs (Etienne et al., 2010), a detailed description of normal structures of the concerned region constitutes a necessary basis for application of an US-guided technique for subarachnoid space puncture. The transversal and sagittal images described in this study have been used successfully for indirect guidance of the needle for cisternal puncture. The US anatomy of the craniocervical junction was already described in the anaesthetized brachycephalic dog by Schmidt et al. (2008). The US features described in the present study were similar to this previous report (Schmidt et al., 2008), although dogs and cats were awake for US reference imaging. A detailed knowledge of the normal US appearance of the atlanto-occipital transition allows the identification of potential local abnormalities such as caudal cerebellar displacement in dogs with a Chiari-like malformation (Schmidt et al., 2008). In humans, the sonography of the craniocervical junction allows assessing of congenital malformation (Chiari II), subarachnoid haemorrhage (Cramer et al., 1986) and distinction between cystic or solid lesions, distinction between an enlarged or atrophic spinal cord (Braun et al., 1983). Recognition of local abnormalities at US is useful before realizing a puncture in the CM as it can potentially avoid fatal complications. This should be considered in dogs with possible cerebellar herniation as in Cavalier King Charles Spaniels (Rusbridge and Knowler, 2004; Cerda-Gonzalez et al., 2009). As demonstrated by Schmidt and co-authors, the pia mater was not visualized as a separated structure, as the hyperechoic lines delineating the spinal cord are due to differences in acoustic impedance between the CSF and the spinal cord (Schmidt et al., 2008).

Cisternal puncture is commonly performed blindly using anatomic landmarks (Oliver et al., 1997; Ducoté and Dewey, 2003). This procedure is technically easier than the lumbar puncture (Oliver et al., 1997) but is considered at higher risk (Lujan Feliu-Pascual et al., 2008; Di Terlizzi and Platt, 2009). This technique is associated with uncommon but significant risks such as direct brainstem trauma, cerebral and/or cerebellar herniation, and central nervous system haemorrhage (Wamsley and Alleman, 2004). The incidence of brain herniation following a CSF collection has been found to be slightly higher in cats than dogs (Ducoté and Dewey, 2003). The present study demonstrated the feasibility of an US-guided technique for CSF puncture in the CM. The goal is not to supplant the usual blind technique but to supply an easily available help for difficult punctures, to reduce unsuccessful attempts and to reduce the incidence of blood contamination.

tion of the CSF which distorts its analysis. A marker could potentially be placed on the spinal needle after depth measurements on the ultrasound images, but this step was not performed in this study. In addition, a statistical study on the correlation between the depth of insertion of the spinal needle and the distance measured on ultrasound images was not the aim of this study.

In humans, the indirect US-guided technique for the lumbar puncture is considered easier to use compared with a non-image-guided technique (Peterson and Abele, 2005; Nomura et al., 2007). The sonographic visualization of the CM allows its exact localization and increases the confidence of the operator. This US-assisted technique can also be useful for fatty patients where the anatomic landmarks are difficult to palpate.

Indirect US guidance when performed using a sagittal approach is used commonly for lumbar subarachnoid space puncture in humans (Peterson and Abele, 2005; Ferre and Sweeney, 2007). In dogs and cats, the approach for the cisternal puncture is also sagittal (Oliver et al., 1997; Ducoté and Dewey, 2003; Tipold, 2003; Elias and Brown, 2008). Because of the small acoustic window available at the craniocervical junction (for the probe and the needle), the indirect US guidance technique was chosen. Direct US guidance may have been feasible using a parasagittal approach to visualize the needle on transverse US images of the CM. Noting the parasagittal position of the lateral venous sinus, the sagittal approach and an indirect US-guided technique were chosen. Moreover, indirect guidance compared with direct US guidance does not require eye-hand coordination to place the needle in the US plane, which is difficult for inexperienced ultrasonographers.

Nevertheless, the success of the indirect US-guided technique lies in the good location of the landmarks with the marker and correct immobilization of the skin to avoid mark displacement. The position of the patient is also crucial as in any technique for cisternal puncture: the nose of the animal should be parallel to the sampling table and its neck must be vigorous flexed (Elias and Brown, 2008) to increase the mid-laminar distance at occiput-C1. As in the blind technique, the midline should be assessed by palpation of the occipital protuberance and the spinous process of the cervical vertebra and not by visual approximation.

Finally, while some veterinarians routinely interpret US images, many others have limited experience and may have more difficulty obtaining and interpreting US images of the occipitocervical junction. However, the relatively simple anatomy of the region and the ease in visualizing the CM suggest that little experience may be sufficient to apply the technique.

This study demonstrates the feasibility of an indirect US-guided cisternal puncture in dogs and cats. Further studies will be interesting to assess the repeatability of this technique at several operators with different experience levels.

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